Desjardins, Guillaume, Simonyan, Karen, Pascanu, Razvan, and Kavukcuoglu, Koray. Natural neural networks. *NIPS*, 2015.

Dinh, Laurent, Krueger, David, and Bengio, Yoshua. Nice: Nonlinear independent components estimation. *arXiv preprint arXiv:1410.8516*, 2014.

Dinh, Laurent, Sohl-Dickstein, Jascha, and Bengio, Samy. Density estimation using real nvp. In *ICLR'2017, arXiv:1605.08803*, 2016.

Duchi, John, Hazan, Elad, and Singer, Yoram. Adaptive subgradient methods for online learning and stochastic optimization. *Journal of Machine Learning Research*, 12(Jul):2121–2159, 2011.

Gehring, Jonas, Auli, Michael, Grangier, David, and Dauphin, Yann N. A convolutional encoder model for neural machine translation. *arXiv preprint arXiv:1611.02344*, 2016.

Glorot, Xavier, Bordes, Antoine, and Bengio, Yoshua. Deep sparse rectifier neural networks. In *Aistats*, volume 15, pp. 275, 2011.

Gonen, Alon and Shalev-Shwartz, Shai. Fast rates for empirical risk minimization of strict saddle problems. *arXiv preprint arXiv:1701.04271*, 2017.

Goodfellow, Ian J, Warde-Farley, David, Mirza, Mehdi, Courville, Aaron C, and Bengio, Yoshua. Maxout networks. *ICML (3)*, 28: 1319–1327, 2013.

Goodfellow, Ian J, Shlens, Jonathon, and Szegedy, Christian. Explaining and harnessing adversarial examples. In *ICLR'2015 arXiv:1412.6572*, 2015.

Graves, Alex, Mohamed, Abdel-rahman, and Hinton, Geoffrey. Speech recognition with deep recurrent neural networks. In *Acoustics, speech and signal processing (icassp), 2013 ieee international conference on*, pp. 6645–6649. IEEE, 2013.

Hannun, Awni Y., Case, Carl, Casper, Jared, Catanzaro, Bryan, Diamos, Greg, Elsen, Erich, Prenger, Ryan, Satheesh, Sanjeev, Sengupta, Shubho, Coates, Adam, and Ng, Andrew Y. Deep speech: Scaling up end-to-end speech recognition. *CoRR*, abs/1412.5567, 2014. URL http://arxiv.org/abs/1412.5567.

Hardt, Moritz, Recht, Ben, and Singer, Yoram. Train faster, generalize better: Stability of stochastic gradient descent. In Balcan, Maria-Florina and Weinberger, Kilian Q. (eds.), *Proceedings of the 33nd International Conference on Machine Learning, ICML 2016, New York City, NY, USA, June 19-24, 2016*, volume 48 of *JMLR Workshop and Conference Proceedings*, pp. 1225–1234. JMLR.org, 2016. URL http://jmlr.org/proceedings/papers/v48/hardt16.html.

He, Kaiming, Zhang, Xiangyu, Ren, Shaoqing, and Sun, Jian. Delving deep into rectifiers: Surpassing human-level performance on imagenet classification. In *Proceedings of the IEEE international conference on computer vision*, pp. 1026–1034, 2015.

He, Kaiming, Zhang, Xiangyu, Ren, Shaoqing, and Sun, Jian. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 770–778, 2016.

Hinton, Geoffrey E and Van Camp, Drew. Keeping the neural networks simple by minimizing the description length of the weights. In *Proceedings of the sixth annual conference on Computational learning theory*, pp. 5–13. ACM, 1993.

Hochreiter, Sepp and Schmidhuber, Jürgen. Flat minima. *Neural Computation*, 9(1):1–42, 1997.

Hyvärinen, Aapo and Pajunen, Petteri. Nonlinear independent component analysis: Existence and uniqueness results. *Neural Networks*, 12(3):429–439, 1999.

Im, Daniel Jiwoong, Tao, Michael, and Branson, Kristin. An empirical analysis of deep network loss surfaces. *arXiv preprint arXiv:1612.04010*, 2016.

Ioffe, Sergey and Szegedy, Christian. Batch normalization: Accelerating deep network training by reducing internal covariate shift. In Bach & Blei (2015), pp. 448–456. URL http://jmlr.org/proceedings/papers/v37/ioffe15.html.

Jarrett, Kevin, Kavukcuoglu, Koray, LeCun, Yann, et al. What is the best multi-stage architecture for object recognition? In *Computer Vision, 2009 IEEE 12th International Conference on*, pp. 2146–2153. IEEE, 2009.

Keskar, Nitish Shirish, Mudigere, Dheevatsa, Nocedal, Jorge, Smelyanskiy, Mikhail, and Tang, Ping Tak Peter. On large-batch training for deep learning: Generalization gap and sharp minima. In *ICLR'2017, arXiv:1609.04836*, 2017.

Kingma, Diederik P, Salimans, Tim, Jozefowicz, Rafal, Chen, Xi, Sutskever, Ilya, and Welling, Max. Improved variational inference with inverse autoregressive flow. In Lee, D. D., Sugiyama, M., Luxburg, U. V., Guyon, I., and Garnett, R. (eds.), *Advances in Neural Information Processing Systems 29*, pp. 4743–4751. Curran Associates, Inc., 2016.

Klyachko, Alexander A. Random walks on symmetric spaces and inequalities for matrix spectra. *Linear Algebra and its Applications*, 319(1-3):37–59, 2000.

Krizhevsky, Alex, Sutskever, Ilya, and Hinton, Geoffrey E. Imagenet classification with deep convolutional neural networks. In *Advances in neural information processing systems*, pp. 1097–1105, 2012.

Lafond, Jean, Vasilache, Nicolas, and Bottou, Léon. About diagonal rescaling applied to neural nets. *ICML Workshop on Optimization Methods for the Next Generation of Machine Learning*, 2016.

Larsen, Anders Boesen Lindbo, Sønderby, Søren Kaae, and Winther, Ole. Autoencoding beyond pixels using a learned similarity metric. *CoRR*, abs/1512.09300, 2015. URL http://arxiv.org/abs/1512.09300.

Montufar, Guido F, Pascanu, Razvan, Cho, Kyunghyun, and Bengio, Yoshua. On the number of linear regions of deep neural networks. In *Advances in neural information processing systems*, pp. 2924–2932, 2014.

Nair, Vinod and Hinton, Geoffrey E. Rectified linear units improve restricted boltzmann machines. In *Proceedings of the 27th international conference on machine learning (ICML-10)*, pp. 807–814, 2010.

Nesterov, Yurii and Vial, Jean-Philippe. Confidence level solutions for stochastic programming. *Automatica*, 44(6):1559–1568, 2008.

Neyshabur, Behnam, Salakhutdinov, Ruslan R, and Srebro, Nati. Path-sgd: Path-normalized optimization in deep neural networks. In *Advances in Neural Information Processing Systems*, pp. 2422–2430, 2015.

Pascanu, Razvan and Bengio, Yoshua. Revisiting natural gradient for deep networks. *ICLR*, 2014.

Raghu, Maithra, Poole, Ben, Kleinberg, Jon, Ganguli, Surya, and Sohl-Dickstein, Jascha. On the expressive power of deep neural networks. *arXiv preprint arXiv:1606.05336*, 2016.

Rezende, Danilo Jimenez and Mohamed, Shakir. Variational inference with normalizing flows. In Bach & Blei (2015), pp. 1530–1538. URL http://jmlr.org/proceedings/papers/v37/rezende15.html.

Sagun, Levent, Bottou, Léon, and LeCun, Yann. Singularity of the hessian in deep learning. *arXiv preprint arXiv:1611.07476*, 2016.

Salimans, Tim and Kingma, Diederik P. Weight normalization: A simple reparameterization to accelerate training of deep neural networks. In *Advances in Neural Information Processing Systems*, pp. 901–901, 2016.

Salimans, Tim, Goodfellow, Ian, Zaremba, Wojciech, Cheung, Vicki, Radford, Alec, and Chen, Xi. Improved techniques for training gans. In *Advances in Neural Information Processing Systems*, pp. 2226–2234, 2016.

Saxe, Andrew M., McClelland, James L., and Ganguli, Surya. Exact solutions to the nonlinear dynamics of learning in deep linear neural networks. *CoRR*, abs/1312.6120, 2013. URL http://arxiv.org/abs/1312.6120.

Simonyan, Karen and Zisserman, Andrew. Very deep convolutional networks for large-scale image recognition. In *ICLR'2015, arXiv:1409.1556*, 2015.

Sutskever, Ilya, Vinyals, Oriol, and Le, Quoc V. Sequence to sequence learning with neural networks. In *Advances in neural information processing systems*, pp. 3104–3112, 2014.

Swirszcz, Grzegorz, Czarnecki, Wojciech Marian, and Pascanu, Razvan. Local minima in training of deep networks. *CoRR*, abs/1611.06310, 2016.

Szegedy, Christian, Zaremba, Wojciech, Sutskever, Ilya, Bruna, Joan, Erhan, Dumitru, Goodfellow, Ian, and Fergus, Rob. Intriguing properties of neural networks. In *ICLR'2014, arXiv:1312.6199*, 2014.

Szegedy, Christian, Liu, Wei, Jia, Yangqing, Sermanet, Pierre, Reed, Scott, Anguelov, Dragomir, Erhan, Dumitru, Vanhoucke, Vincent, and Rabinovich, Andrew. Going deeper with convolutions. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 1–9, 2015.

Theis, Lucas, Oord, Aäron van den, and Bethge, Matthias. A note on the evaluation of generative models. In *ICLR'2016 arXiv:1511.01844*, 2016.

Wu, Yonghui, Schuster, Mike, Chen, Zhifeng, Le, Quoc V, Norouzi, Mohammad, Macherey, Wolfgang, Krikun, Maxim, Cao, Yuan, Gao, Qin, Macherey, Klaus, et al. Google's neural machine translation system: Bridging the gap between human and machine translation. *arXiv preprint arXiv:1609.08144*, 2016.

Zhang, Chiyuan, Bengio, Samy, Hardt, Moritz, Recht, Benjamin, and Vinyals, Oriol. Understanding deep learning requires rethinking generalization. In *ICLR'2017, arXiv:1611.03530*, 2017.

A Radial transformations

We show an elementary transformation to locally perturb the geometry of a finite-dimensional vector space and therefore affect the relative flatness between a finite number minima, at least in terms of spectral norm of the Hessian. We define the function:

$$\begin{aligned}
&\forall \delta > 0, \forall \rho \in]0,\delta[, \forall (r,\hat{r}) \in \mathbb{R}_+ \times]0,\delta[, \\
&\psi(r,\hat{r},\delta,\rho) = \mathbb{1}\big(r \notin [0,\delta]\big)\, r + \mathbb{1}\big(r \in [0,\hat{r}]\big)\, \rho\, \frac{r}{\hat{r}} \\
&\qquad + \mathbb{1}\big(r \in]\hat{r},\delta]\big) \left((\rho - \delta)\, \frac{r-\delta}{\hat{r}-\delta} + \delta \right) \\
&\psi'(r,\hat{r},\delta,\rho) = \mathbb{1}\big(r \notin [0,\delta]\big) + \mathbb{1}\big(r \in [0,\hat{r}]\big)\, \frac{\rho}{\hat{r}} \\
&\qquad + \mathbb{1}\big(r \in]\hat{r},\delta]\big)\, \frac{\rho-\delta}{\hat{r}-\delta}
\end{aligned}$$

For a parameter $\hat{\theta} \in \Theta$ and $\delta > 0, \rho \in]0,\delta[, \hat{r} \in]0,\delta[$, inspired by the *radial flows* (Rezende & Mohamed, 2015) in we can define the *radial transformations*

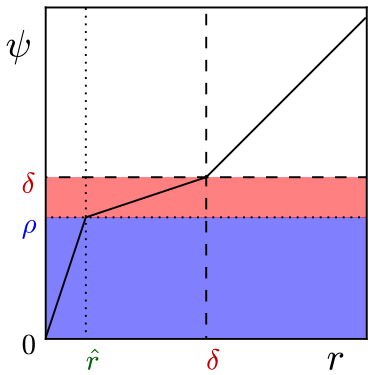
$$\forall \theta \in \Theta,\ g^{-1}(\theta) = \frac{\psi\Big(\|\theta - \hat{\theta}\|_2, \hat{r}, \delta, \rho\Big)}{\|\theta - \hat{\theta}\|_2} \left(\theta - \hat{\theta}\right) + \hat{\theta}$$

with Jacobian

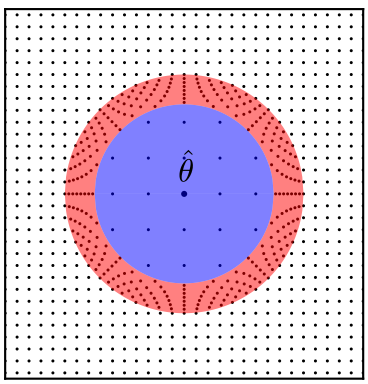
$$\begin{aligned}
\forall \theta \in \Theta,\ (\nabla g^{-1})(\theta) = &\ \psi'(r,\hat{r},\delta,\rho)\, \mathbb{I}_n \\
&- \mathbb{1}\big(r \in]\hat{r},\delta]\big) \frac{\delta(\hat{r}-\rho)}{r^3(\hat{r}-\delta)}\, (\theta - \hat{\theta})^T (\theta - \hat{\theta}) \\
&+ \mathbb{1}\big(r \in]\hat{r},\delta]\big) \frac{\delta(\hat{r}-\rho)}{r(\hat{r}-\delta)}\, \mathbb{I}_n,
\end{aligned}$$

with $r = \|\theta - \hat{\theta}\|_2$.

First, we can observe in Figure 6 that these transformations are purely local: they only have an effect inside the ball $B_2(\hat{\theta}, \delta)$. Through these transformations, you can arbitrarily perturb the ranking between several minima in terms of flatness as described in Subsection 5.1.



(a) $\psi(r, \hat{r}, \delta, \rho)$



(b) $g^{-1}(\theta)$

Figure 6: An example of a radial transformation on a 2-dimensional space. We can see that only the area in blue and red, i.e. inside $B_2(\hat{\theta}, \delta)$, are affected. Best seen with colors.

B Considering the bias parameter

When we consider the bias parameter for a one (hidden) layer neural network, the non-negative homogeneity property translates into

$$\begin{aligned} y &= \phi_{rect}(x \cdot \theta_1 + b_1) \cdot \theta_2 + b_2 \\ &= \phi_{rect}(x \cdot \alpha\theta_1 + \alpha b_1) \cdot \alpha^{-1}\theta_2 + b_2, \end{aligned}$$

which results in conclusions similar to section 4.

For a deeper rectified neural network, this property results in

$$\begin{aligned} y &= \phi_{rect}\Big(\phi_{rect}\big(\cdots \phi_{rect}(x \cdot \theta_1 + b_1) \cdots\big) \cdot \theta_{K-1} + b_{K-1}\Big) \\ &\quad \cdot \theta_K + b_K \\ &= \phi_{rect}\Big(\phi_{rect}\big(\cdots \phi_{rect}(x \cdot \alpha_1\theta_1 + \alpha_1 b_1) \cdots\big) \\ &\quad \cdot \alpha_{K-1}\theta_{K-1} + \prod_{k=1}^{K-1} \alpha_k b_{K-1}\Big) \cdot \alpha_K\theta_K + b_K \end{aligned}$$

for $\prod_{k=1}^{K} \alpha_k = 1$. This can decrease the amount of eigenvalues of the Hessian that can be arbitrarily influenced.

C Rectified neural network and Lipschitz continuity

Relative to recent works (Hardt et al., 2016; Gonen & Shalev-Shwartz, 2017) assuming *Lipschitz continuity* of the loss function to derive uniform stability bound, we make the following observation:

Theorem 6. *For a one-hidden layer rectified neural network of the form*

$$y = \phi_{rect}(x \cdot \theta_1) \cdot \theta_2,$$

if L is not constant, then it is not Lipschitz continuous.

Proof. Since a Lipschitz function is necessarily absolutely continuous, we will consider the cases where L is absolutely continuous. First, if L has zero gradient almost everywhere, then L is constant.

Now, if there is a point θ with non-zero gradient, then by writing

$$\begin{aligned} (\nabla L)(\theta_1, \theta_2) = [&(\nabla_{\theta_1} L)(\theta_1, \theta_2) \\ &(\nabla_{\theta_2} L)(\theta_1, \theta_2)], \end{aligned}$$

we have

$$\begin{aligned} (\nabla L)(\alpha\theta_1, \alpha^{-1}\theta_2) = [&\alpha^{-1}(\nabla_{\theta_1} L)(\theta_1, \theta_2) \\ &\alpha(\nabla_{\theta_2} L)(\theta_1, \theta_2)]. \end{aligned}$$

Without loss of generality, we consider $(\nabla_{\theta_1} L)(\theta_1, \theta_2) \neq 0$. Then the limit of the norm

$$\begin{aligned} \|(\nabla L)(\alpha\theta_1, \alpha^{-1}\theta_2)\|_2^2 = \ &\alpha^{-2}\|(\nabla_{\theta_1} L)(\theta_1, \theta_2)\|_2^2 \\ &+ \alpha^2\|(\nabla_{\theta_2} L)(\theta_1, \theta_2)\|_2^2 \end{aligned}$$

of the gradient goes to $+\infty$ as α goes to 0. Therefore, L is not Lipschitz continuous. □

This result can be generalized to several other models containing a one-hidden layer rectified neural network, including deeper rectified networks.